Debug mate

Teammates

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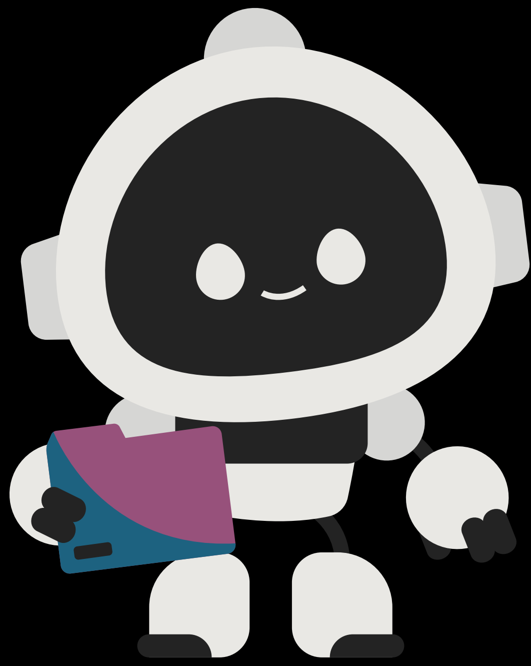
AI Avengers

Problem Statement



1. **Beginners struggle with errors**

   Students often get stuck on syntax errors, logic bugs, and runtime exceptions and find it difficult to understand error messages.

2. **Copying solutions instead of learning**

   Many students copy-paste error messages into search engines or ask others for solutions without understanding why the bug occurred.

3. **Learning gaps and dependency**

   This slows down learning, creates dependency on external help, and leaves gaps in basic programming concepts.

WHY IT MATTERS

Debugging is where real programming understanding is built — or lost.

Objectives

Explain in Programming Language

Detect and explain programming errors in simple, beginner-friendly language.

Localize the Bug

Identify the exact line or logic causing the problem.

Hint Before Revealing

Guide students with a hint before showing the complete solution.

Build Root-Cause Understanding

Help students understand the programming concept behind the bug.

Existing SYSTEM / CURRENT CHALLENGES

DECODING THE COMPLEXITY

- Unravel tangled code structures.
- Translate business logic to efficient code.
- Identify subtle syntax and logical errors.

THE ART OF DEBUGGING

- Isolate issues using breakpoints and traces.
- Profile performance and trace memory leaks.
- Use logs effectively for faster resolution.

BUILDING RESILIENT ARCHITECTURE

- Design patterns for scalability.
- Abstract components for maintainability.
- Create comprehensive unit and integration tests.

CONTINUOUS IMPROVEMENT

- Conduct thorough, constructive code reviews.
- Apply refactoring best practices.
- Automate code quality checks.

Our Solution

DebugMate: A Staged AI Tutor

DebugMate takes code and error as input, analyzes the problem using Generative AI and guides students to understand and fix bugs step-by-step.

GENERATIVE AI ANALYSIS

AI analyzes the code, detects the bug and prepares personalized guidance.

01 HINT

Guide the student toward the bug without revealing the solution.

02 CONCEPT

Explain the programming concept behind the error in simple language.

03 FIX

Show the corrected code and explain what was changed.

HOW IT WORKS

Code + Error (Input) → AI Analysis (Detect & Understand) → Guided Debugging (Hint → Concept → Fix) → Student Learning (Understand & Improve)

Generative AI adapts to each student's code and error, making DebugMate more than just an error fixer – it's a learning partner.

Tools & Technologies

Technologies powering DebugMate

PLATFORM

Replit AI

Replit Agent /
AI coding environment

LANGUAGES

Python / JavaScript

Choose the language
used in your demo

REPLIT AI

Powering DebugMate
with Generative AI

AI MODEL

Integrated LLM

Replit-integrated
Generative AI / LLM

CODE EXECUTION

Replit Runtime

Code execution sandbox
to run and test code

WEB INTERFACE

HTML / CSS / JavaScript

Simple web UI for code input,
error input and AI response

DEMO

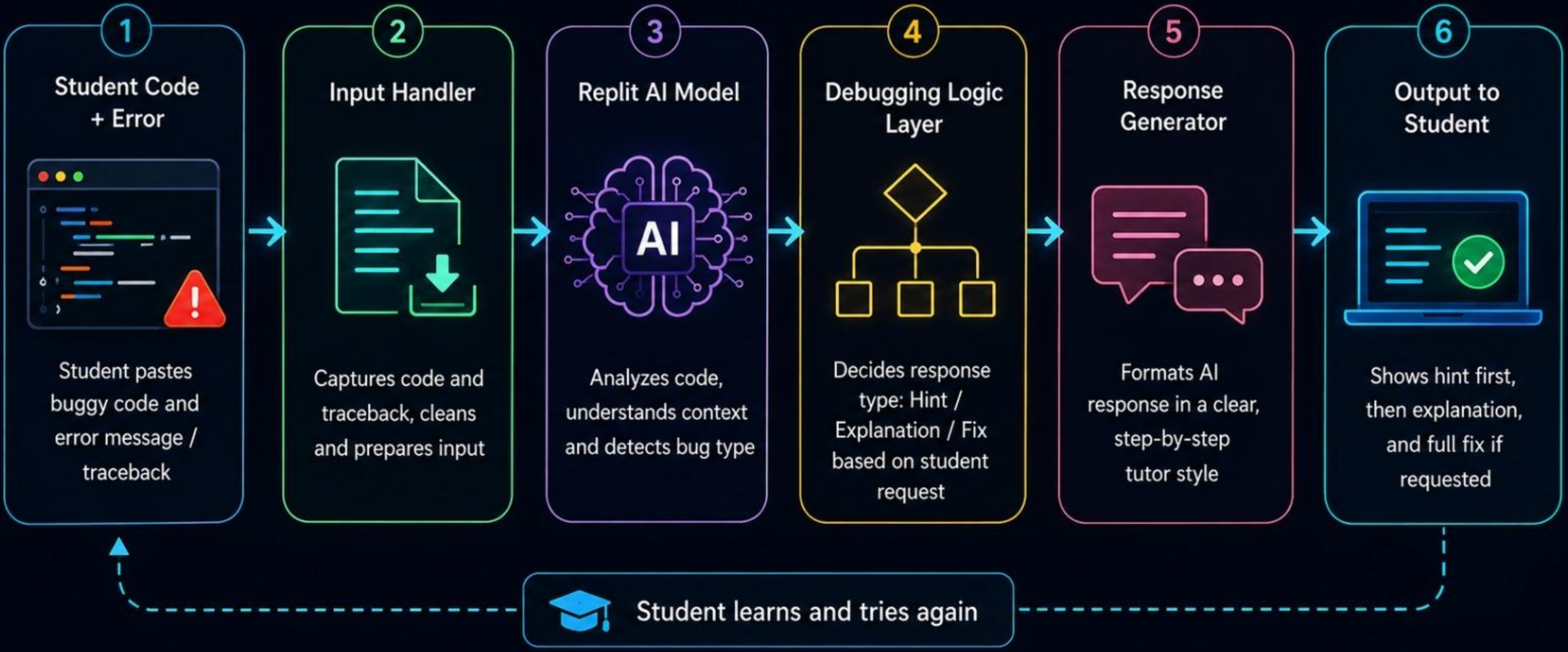
Programming Languages

Python | C | C++ | Java | HTML | JavaScript

Together, these technologies power DebugMate's AI-assisted debugging experience.

ARCHITECTURE / WORKFLOW

How DebugMate Works



Key Features

🐞 Error Explanation

Converts complex error messages into simple, beginner-friendly language.

💡 Hint-First Mode

Provides a hint before the solution, encouraging students to think independently.

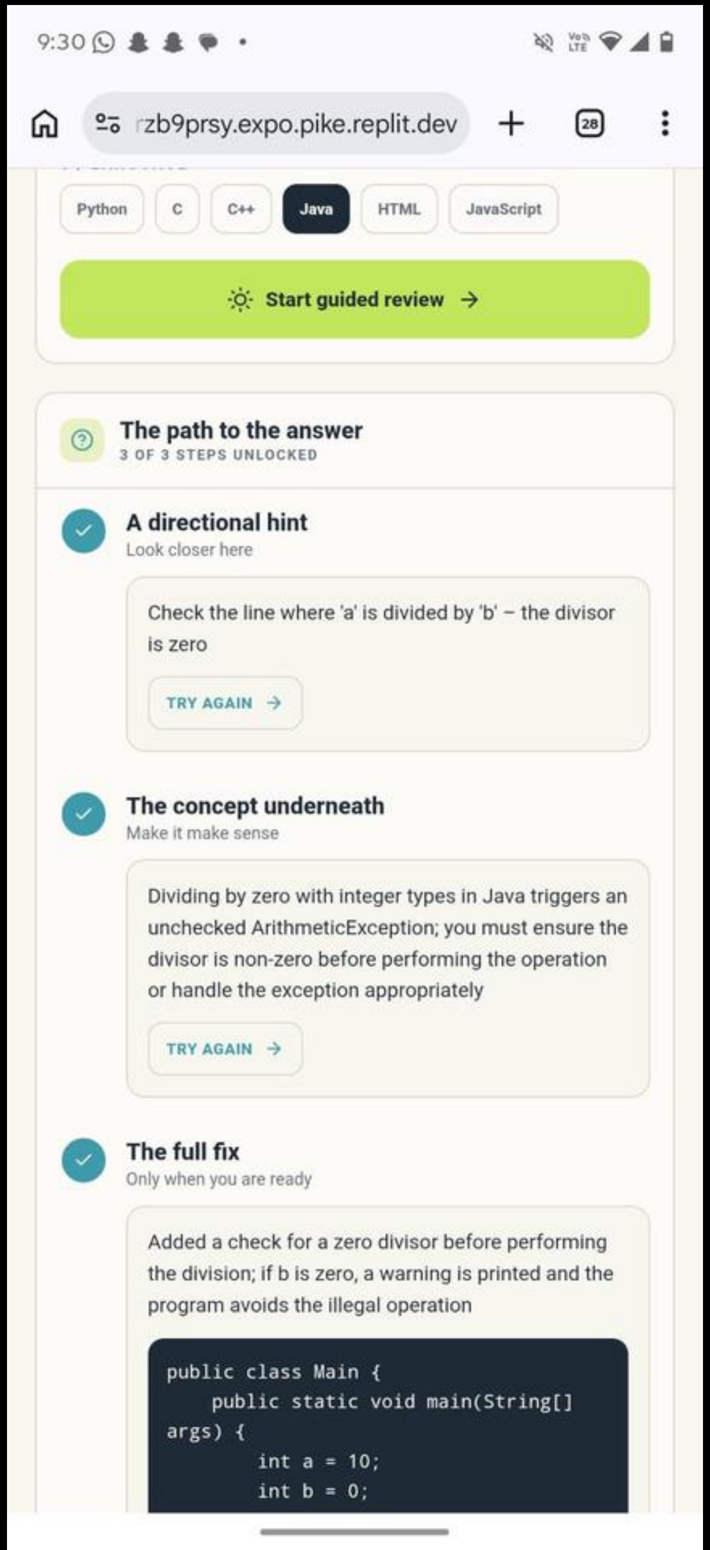
📍 Bug Localization

Identifies the specific line, statement, or logic causing the problem.

🔧 Full-Fix Mode

Provides the corrected code on request and explains what was changed.

Demo/Screenshots



Test Case

TEST CASE SUMMARY

Division by Zero (All Languages)

PASS

TEST CASE ID	BUG TYPE	LANGUAGES	OBJECTIVE	DATE
TC-DZ-001	Division by Zero	Python, C, C++, Java, HTML, JavaScript	Detect and handle division by zero and provide a fix.	25 May 2025

Language	Buggy Code	Issue / Error	Expected Result	Fix / Corrected Code	Status

Python	x = 10 / 0
	print(x)
		ZeroDivisionError: division by zero	Handle zero divisor and avoid runtime error.	x, y = 10, 0
				y = y if y != 0 else None
				if y is None:
				print('Error: Division
				b zero')
				else:
				print(x / y)
					PASS

C	#include <stdio.h>
	int main(){
	printf("%d", 10 / 0);
	return 0;
	}
		Runtime Error (Illegal division by zero)	Ensure divisor is non-zero before performing division.	#include <stdio.h>
				int main(){
				int b = 2; // non-zero
				printf("%d", 10 / b);
				return 0;
				}
					PASS

C++	#include <iostream>
	using namespace std;
	int main() {
	cout << 10 / 0;
	return 0;
	}
		Runtime Error (Illegal division by zero / Crash)	Use a non-zero denominator or validate before division.	#include <iostream>
				using namespace std;
				int main() {
				int d = 2; // non-zero
				cout << 10 / d;
				return 0;
				}
					PASS

Java	public class Main {
	public static void main
	(String[] args){
	System.out.println(10/0);
	}
		ArithmeticException: / by zero	Check denominator before performing division.	public class Main {
				public static void main
				(String[] args) {
				int d == 2; // non-zero
				System.out.println(10 / d);
				}
					PASS

HTML	<html> <h1>Welcome</h1> <p>This is a paragraph<p> </html>	Unclosed <p> tag (Malformed HTML)	Close all opened tags properly.	<!DOCTYPE html> <html> <head><title>Welcome Page</title></head> <body> Welcome This is a paragraph </body> </html>	PASS

VALIDATION CHECKLIST

- Bug identified
- Hint generated
- Concept explained
- Fix provided
- Fix verified

ALL TEST CASES PASSED

Demo Link

https://udify.app/workflow/LdABxlwCMHhxiPDi

https://be93932b-1c5c-4627-923f-8e994cb446f2-00-3jp39rzb9prsy.expo.pike.replit.dev/?nativeBrowserPresentationStyle=fullScreen

ADVANTAGES & DISADVANTAGES

ADVANTAGES

01. ACTIVE LEARNING
Encourages students to understand and solve bugs instead of copy-pasting solutions.

02. INSTANT FEEDBACK
Provides debugging guidance anytime without waiting for a mentor.

03. BEGINNER FRIENDLY
Explains complex error messages in simple, easy-to-understand language.

04 - PERSONALIZED GUIDANCE
Analyzes the student's specific code and provides customized explanations.

05 - MULTIPLE ERROR TYPES
Can assist with syntax, runtime, and logical errors.

DISADVANTAGES

01 - LARGE CODEBASES
May struggle to analyze very large or multi-file projects.

02 - AI ACCURACY
Responses depend on the reasoning and accuracy of the underlying AI model.

03. NOT A TEACHER REPLACEMENT
Students still need hands-on programming and debugging practice.

04 - INTERNET DEPENDENCY
Requires internet / API access for AI-powered functionality.

Future Enhancements

👁️ Visual Debugger

Add a step-by-step debugger that visually shows variable values and program flow.

📊 Personalized Learning Analytics

Track students' progress and common error patterns to provide personalized guidance.

🎙️ Voice-Based Assistance

Add voice explanations to make debugging more accessible and interactive.

💻 IDE Integration

Extend DebugMate beyond Replit with a VS Code extension and other development environments.

Conclusion

DebugMate transforms debugging into a learning opportunity by using Replit AI to understand code contextually, explain concepts, and guide students toward the solution. It addresses the gap in programming education by teaching students how to debug rather than simply fixing their bugs for them.


THANK
YOU